

Quiz 2

Course: CSE422

Time: 30 minutes

Student ID: _____

Section: _____

Magical Diagnosis Problem

At **Hogwarts Hospital Wing**, a student (Mr. X) shows symptoms of a rare magical infection called *Confundus Curse*.

Madam Pomfrey performs a diagnostic spell.

The following is known about the spell:

- If a student is truly cursed, the spell detects it correctly with probability **0.92**.
- If a student is not cursed, the spell incorrectly signals a curse with probability **0.15**.
- Historically, **17%** of students visiting the hospital wing are actually cursed.

Mr. X's **first spell result is positive** (indicating a curse).

Worried, Madam Pomfrey repeats the spell **two more times**, and both results come back **negative**.

Assume that the spell outcomes are **conditionally independent** given whether the student is truly cursed or not.

- (a) Using only the **first positive result**, compute the probability that Mr. X is actually **not cursed**. [6]
- (b) Using **all three spell results** (one positive followed by two negatives), determine which is more likely:

Mr. X is actually cursed or **Mr. X is not cursed**

Clearly justify your answer using probability calculations.

[8]

- (c) Explain why assuming conditional independence of repeated spell results leads to a certain probability model. **Name** the Model and briefly discuss one practical reason why this assumption might fail in real-world (or magical) diagnostic systems. [6]
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